

CONSTRUCTING AND CHARACTERIZING FIBER OPTIC SYSTEMS: MULTIPLEXING AND DIAMETER VARIABILITY

ALEX SOTO

ABSTRACT

Fiber optics has revolutionized modern communication systems. Transmitting information as pulses of light through optical fibers has enabled high-speed, high-bandwidth, and reliable communication across the globe (Paschotta). In this study, we explore the construction and characterization of fiber optic systems with a single mode fiber ($\sim 10\text{ }\mu\text{m}$ in diameter) and multimode fiber ($\sim 50\text{ }\mu\text{m}$ in diameter). Multiplexing was employed to transmit three distinct wavelengths of light, 405 nm, 532 nm, and 650 nm, across the fibers. When necessary, a diffraction grating was used to demux the emitted light. Images were captured at the emitting ends of the fibers to assess the behavior of each color. In addition, the efficiency of the systems was evaluated by measuring the generated photocurrent at both ends of the fiber. Ultimately, we built four fiber optic systems (one multi-mode and three single-mode) and characterized each of them qualitatively by their outputted images and quantitatively by their efficiencies.

INTRODUCTION

Fiber optics has enabled the establishment of large communication systems including telephone systems, the internet, cable TV, etc. (Paschotta). Compared to traditional copper wires, fiber optics has significant advantages. In addition to higher speeds and greater reliability, fiber optics enables the use of wavelength division multiplexing—where different wavelengths of light are sent down the same fiber (Paschotta). This increases the throughput of the system as more information can be sent in parallel.

Regarding multiplexing, there exists a tradeoff between the use of single-mode ($\sim 10\text{ }\mu\text{m}$ in diameter) and multi-mode ($\sim 50\text{ }\mu\text{m}$ in diameter) fibers. Single-mode fibers, due to their narrow core, have no modal dispersion since light can only travel in one mode (Congreve). Although

they are more accurate and support higher bandwidth, they are more expensive to install (Paschotta). On the other hand, multi-mode fibers introduce the effects of modal dispersion, which makes them less reliable over long distances, but they are less expensive to install (Congreve, Paschotta).

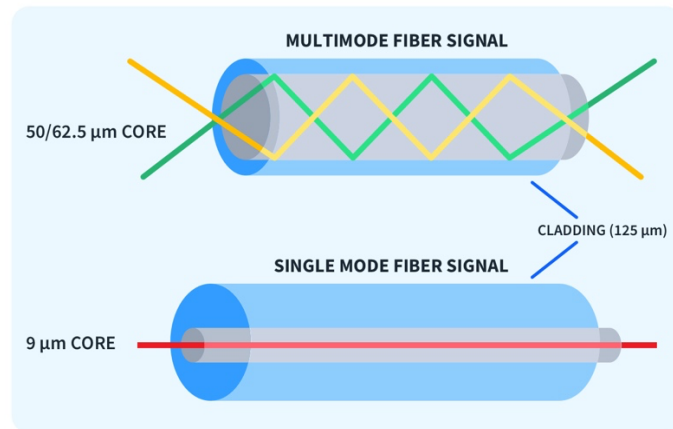


Figure 1: Single-Mode vs Multi-Mode

The purpose of this study is to characterize and assess the performance of fiber optic systems with single-mode and multi-mode fibers. The goal is to find a system that would be most suitable for building a simple communication system in the future.

METHODS

Constructing and Characterizing the Multi-Mode Fiber System:

The ends of the multi-mode fiber were fixed using magnetic mounts. Afterwards, a 200 mm lens was placed a focal length away from one end of the thick fiber. Two 50 mm lenses were placed a focal length away from the same end of the fiber but at an angle. A blue laser was shined onto the 200 mm lens, and red and green lasers were shined onto the other lenses. The use of lenses of different focal lengths served to minimize the angles of the outer lasers. Each laser was carefully aligned until a significant amount of light was emitted from the other end of the fiber.

Once the system was aligned, a 25 mm lens was placed at the emitting end of the fiber. A diffraction grating was placed shortly after, and a camera was placed a focal length away from the lens. The camera was slightly adjusted until the three distinct wavelengths of light could be seen.

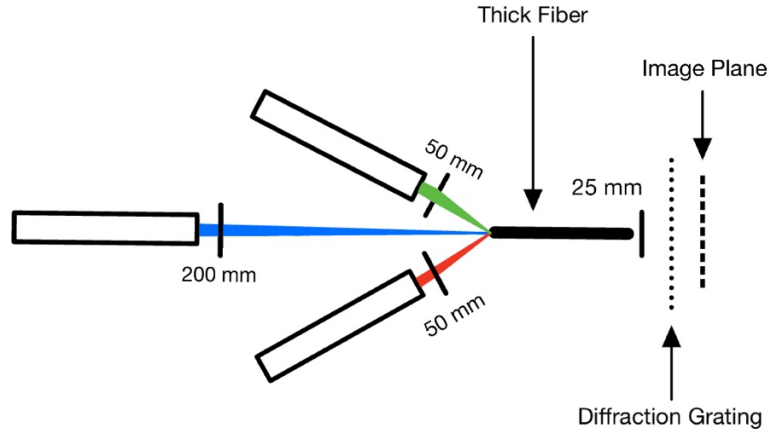


Figure 2: Multi-Mode Fiber System

To assess the efficiency of the system, a photodiode connected to an ammeter was used to measure the generated photocurrent at both ends of the fiber. The photodiode was manually held so that the light directly hit the center, and the highest measurement was recorded. These measurements were taken for each individual laser and for all three simultaneously. The efficiency was calculated as the ratio between the optical power entering the fiber (P_{in}) and the optical power exiting the fiber (P_{out}) (Gamry Instruments). This is equivalent to the ratio between the generated photocurrent at the entrance of the fiber (I_{in}) and the generated photocurrent at the emitting end of the fiber (I_{out}) (Gamry Instruments).

Constructing and Characterizing the Single-Mode Fiber Systems:

Building off the previous multi-mode fiber system, the single-mode fiber system was constructed by replacing the multi-mode fiber with a single-mode fiber and removing the diffracting grating. The lasers were precisely aligned again until a significant amount of light was emitted at the other end of the fiber.

The images were captured and the photocurrents were measured in the same manner. This process was repeated for a total of three trials to study the effects of altering the positions of the lasers.

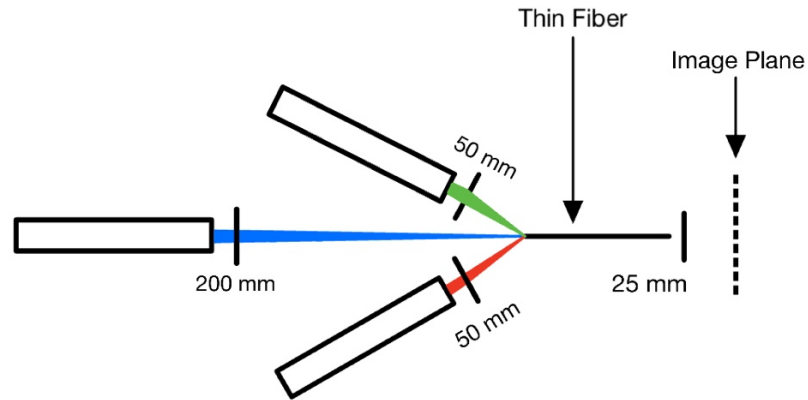


Figure 3: Single-Mode Fiber System

RESULTS

Multi-Mode Fiber System:

The multi-mode fiber system outputted a very white light, which the diffracting grating separated into three distinct wavelengths. The efficiency of the blue centered light was the highest, and the angled colors performed at about half that efficiency.



Figure 4: Thick Fiber Output Image

Laser	Wavelength (nm)	I_{In} (mA)	I_{Out} (mA)	Efficiency (%)
Blue (center)	405	964	112	11.62
Green (angle)	532	2400	128	5.33
Red (angle)	650	1360	78.5	5.77
Cumulative	-	3700	316	8.54

Table 1: Multi-Mode Fiber System Efficiency Data

Single-Mode Fiber System with Centered Blue Laser:

The red and green light was emitted by the single-mode fiber as circles of different radii, while the blue light was emitted as a spot. There was no need to use a diffracting grating for the single-mode fiber due to the skewed ray behavior of the red and green light (Bahaa E. A. Saleh, Malvin Carl Tiech). In the image, the blue light can be seen in the center, while the red and green light can be seen forming a circle with tangent rays. The efficiency of the blue light was significantly higher than the other wavelengths.

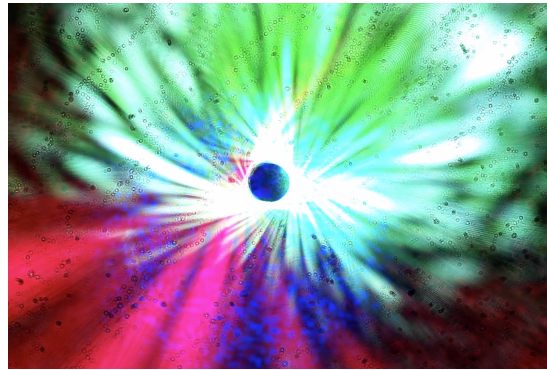


Figure 5: Single-Mode Fiber System with Centered Blue Laser Output Image

Laser	Wavelength (nm)	I_{In} (mA)	I_{Out} (mA)	Efficiency (%)
Blue (center)	405	964	120	12.45
Green (angle)	532	2400	23.6	0.98
Red (angle)	650	1360	27	1.99
Cumulative	-	3700	144	3.89

Table 2: Single-Mode Fiber System with Centered Blue Laser Efficiency Data

Single-Mode Fiber System with Centered Green Laser:

When in the center position, the green light emitted as a circle rather than a spot. The other two colors were also emitted as circles of different radii. In the output image, the red light overpowered. Very faint blue and green circles can be seen. The green light achieved the highest efficiency. Though, it was not as high as the center lasers from the other trials.

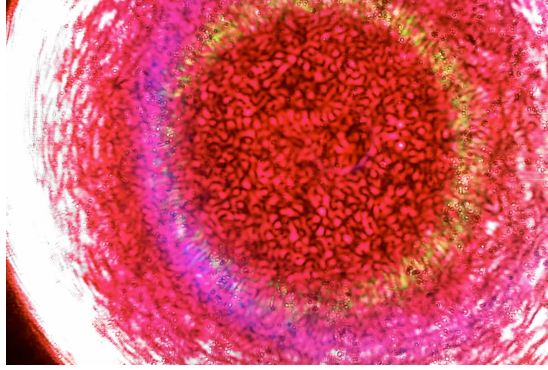


Figure 6: Single-Mode Fiber System with Centered Green Laser Output Image

Laser	Wavelength (nm)	I_{In} (mA)	I_{Out} (mA)	Efficiency (%)
Blue (angle)	405	964	14	1.45
Green (center)	532	2400	140	5.83
Red (angle)	650	1360	27	1.99
Cumulative	-	3700	174	4.70

Table 3: Single-Mode Fiber System with Centered Green Laser Efficiency Data

Single-Mode Fiber System with Centered Red Laser:

The red light, when in the center position, was also emitted as a circle, which can be seen in the outskirts of the output image. The green and blue light behaved as skewed rays (Bahaa E. A. Saleh, Malvin Carl Tiech). The red achieved a significantly higher efficiency. Although the efficiencies of the green and blue light are somewhat abysmal, they appear clearly in the output image.

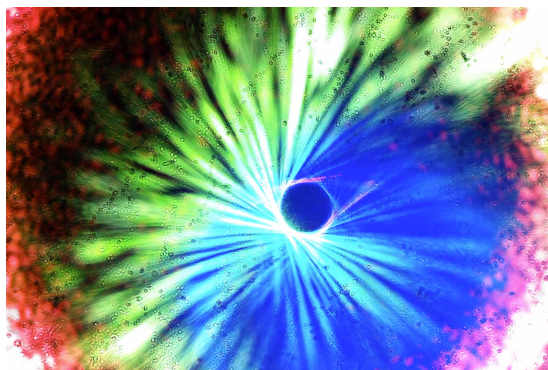


Figure 7: Single-Mode Fiber System with Centered Red Laser Output Image

Laser	Wavelength (nm)	I_{In} (mA)	I_{Out} (mA)	Efficiency (%)
Blue (angle)	405	964	7	0.73
Green (angle)	532	2400	2	0.08
Red (center)	650	1360	210	15.44
Cumulative	-	3700	220	5.95

Table 4: Single-Mode Fiber System with Centered Red Laser Efficiency Data

DISCUSSION

All four fiber optic systems were successful in emitting the three wavelengths of light in an observable manner. The multi-mode fiber system produced very clean images and obtained the highest cumulative efficiency. The single-mode fiber systems produced striking images, but these systems were extremely difficult to align and experienced low efficiencies for the angled lasers. It is also worth noting that the single-mode system with the centered green laser produced an output image that looks different from the other single-mode systems, which could be the result of alignment inconsistencies.

CONCLUSION

The multi-mode system is most suitable for a simple communication system. This system was the easiest to align, the least expensive, and the least sensitive. More work needs to be done to increase the efficiency of the system and alleviate the discrepancy between the centered and angled lasers. Ultimately, the multi-mode system demonstrated its capability to effectively transmit three distinct signals, laying the groundwork for building a short-distanced communication system in the future.

REFERENCES

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